Systematic Classification of the Admissible Field Equations of Gravitation under the Condition of a Spatially Euclidean Static Field

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We give a complete enumeration of the differential expressions of the second order in the fundamental tensor $g_{\mu\nu}$ which can serve as the left-hand member of the field equations of gravitation, subject to three requirements: that the expression be of the second differential order and quadratic at most in the first derivatives, that it transform as a tensor under linear substitutions of the coordinates, and that it be compatible with the spatially Euclidean form of the static field. Of the 75 expressions generated by the enumeration, 28 are shown to be equivalent to others by the identities of Appendix B, leaving 47 distinct candidates. The requirement that the equations pass over into that of Poisson in the Newtonian limit eliminates 31; the requirement that the divergence of the total energy tensor vanish identically eliminates a further 12. Four candidates survive. Of these, three are shown to possess no solution corresponding to the field of a mass point at rest, and the remaining one, K_{28}, is identical with the system of field equations obtained in Ref. [5] by an entirely different route, namely from the law of conservation. We further show that no member of the 47 transforms as a tensor under arbitrary substitutions, so that the demand for general covariance must be abandoned.

I. INTRODUCTION

The search for the field equations of gravitation has hitherto proceeded by trial. A candidate expression is written down, the Newtonian limit is examined, the conservation laws are tested, and if the candidate fails one begins again. The procedure is laborious, and it affords no assurance that the correct expression has not been passed over.

We here replace it by an enumeration. If the class of admissible expressions can be delimited by conditions which are certain, and if the class so delimited is finite, then the correct field equations, if they exist at all, must be among its members, and the search is reduced to the examination of a list.

The delimiting conditions are stated in Sec. II, the enumeration is carried out in Sec. III, and the eliminations occupy Secs. IV and V. The survivors are compared in Sec. VI. Section VII contains what is, in our view, the most consequential result of the investigation: no admissible expression is generally covariant.

II. THE DELIMITING CONDITIONS

A. Differential order

The field equations shall be of the form $E_{\mu\nu} = \kappa\, T_{\mu\nu}$, with $E_{\mu\nu}$ built from $g_{\mu\nu}$ and its first and second derivatives, linear in the second derivatives and at most quadratic in the first. This is the direct analogue of the Poisson equation and requires no defence: equations of higher differential order would require additional initial data for which no physical interpretation is available.

B. Linear covariance

$E_{\mu\nu}$ shall transform as a covariant tensor of the second rank under linear substitutions of the coordinates. The restriction to linear substitutions is discussed in Sec. VII.

C. The spatially Euclidean static field

The static field of a mass point at rest shall be of the form

$$ds^2 = c^2(x,y,z)\, dt^2 - dx^2 - dy^2 - dz^2, \tag{1}$$

that is, the spatial part of the metric shall be Euclidean, and the whole of the gravitational effect shall reside in g_{44}.

This requirement is not one of convenience. It follows from the universality of free fall. In a static field the acceleration of a slowly moving test body is determined by the gradient of g_{44} alone if and only if the spatial components are constant; were g_{11}, g_{22}, g_{33} to vary from place to place, the acceleration would acquire a term depending on the internal constitution of the falling body through the relation between its energy and its stresses, and bodies of different constitution would fall differently. Since the equality of inertial and gravitational mass is verified to a part in 10^9, we must have $\partial_i g_{jk} = 0$ for the spatial components in the static case.[1]

[1] It has been remarked to us that the argument establishes the conclusion for the special case of the homogeneous field produced

III. THE ENUMERATION

Every expression satisfying the conditions of Sec. II is a linear combination of the six structures

$$S^{(1)}_{\mu\nu} = g^{\alpha\beta}\,\partial_\alpha\partial_\beta\,g_{\mu\nu}, \tag{2}$$
$$S^{(2)}_{\mu\nu} = \partial_\mu\partial_\nu \ln\sqrt{-g}, \tag{3}$$
$$S^{(3)}_{\mu\nu} = \partial_\mu\partial^\alpha g_{\alpha\nu} + \partial_\nu\partial^\alpha g_{\alpha\mu}, \tag{4}$$
$$S^{(4)}_{\mu\nu} = g_{\mu\nu}\,\partial^\alpha\partial^\beta g_{\alpha\beta}, \tag{5}$$
$$S^{(5)}_{\mu\nu} = g_{\mu\nu}\,g^{\alpha\beta}\partial_\alpha\partial_\beta \ln\sqrt{-g}, \tag{6}$$
$$S^{(6)}_{\mu\nu} = g^{\alpha\beta}g^{\rho\sigma}\,\partial_\alpha g_{\mu\rho}\,\partial_\beta g_{\nu\sigma}, \tag{7}$$

so that

$$E_{\mu\nu} = \sum_{n=1}^{6} a_n\,S^{(n)}_{\mu\nu}. \tag{8}$$

We normalize $a_1 = 1$, the overall factor being absorbed in κ.

Two relations among the coefficients follow from the requirement that (8) possess the correct tensor weight, namely

$$a_4 = -\tfrac{1}{2}\,(a_1 + a_3), \qquad a_6 = -a_2, \tag{9}$$

the derivation being given in Appendix A. There remain three free coefficients a_2, a_3, a_5.

Restricting these to the set $\{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$, which exhausts the values compatible with the requirement that κ retain its Newtonian dimension, and noting that the first of (9) admits only $a_3 \in \{-1, 0, 1\}$ if a_4 is to lie in the same set, we obtain $5\times 3\times 5 = 75$ expressions. Of these, 28 are equivalent to others in the list by the identities of Appendix B, leaving 47 distinct candidates. They are listed in Table II as $K_1, \ldots, K_{47}$.

IV. ELIMINATION BY THE NEWTONIAN LIMIT

In the weak static field (1), write $g_{44} = c_0^2(1 + 2\varphi/c_0^2)$ with the spatial components constant, and retain first order terms. The 44 component of (8) becomes

$$E_{44} = 2\big(1 + a_3 + 2a_4\big)\,\Delta\varphi + 2\big(a_2 + a_5\big)\,\partial_4\partial_4\varphi + O(\varphi^2). \tag{10}$$

by uniform acceleration, from which it was originally read off, and that the extension to an arbitrary static field is not strictly proved. We have not succeeded in constructing a static field which violates the requirement, and in the absence of such a construction we regard the requirement as established. Should a counterexample be found, the classification below would have to be extended, and the number of admissible expressions would increase considerably.

TABLE I. The four surviving candidates.

	a_2	a_6	Behaviour for the field of a mass point
K_{18}	-1	1	no solution regular at infinity
K_{21}	$-\frac{1}{2}$	$\frac{1}{2}$	solution not asymptotically Minkowskian
K_{28}	$\frac{1}{2}$	$-\frac{1}{2}$	regular, asymptotically Minkowskian
K_{31}	1	-1	solution has g_{44} increasing with r

In the static case the second term vanishes identically, and the equation reduces to that of Poisson provided $1 + a_3 + 2a_4 \neq 0$. By the first of (9), however,

$$1 + a_3 + 2a_4 = 1 + a_3 - (1 + a_3) = 0 \tag{11}$$

identically, so that the leading term cancels and the Newtonian limit is reached only through the next order. Carrying the expansion one step further one finds that the coefficient of $\Delta\varphi$ in the surviving term is proportional to a_3, so that the Newtonian limit exists if and only if

$$a_3 = 0. \tag{12}$$

This eliminates 31 of the 47 candidates, marked I in Table II.

V. ELIMINATION BY THE CONSERVATION LAW

The field equations must be compatible with

$$\partial_\nu\big[\sqrt{-g}\,(\mathfrak{T}^\nu_\mu + \mathfrak{t}^\nu_\mu)\big] = 0 \tag{13}$$

identically, $\mathfrak{t}^\nu_\mu$ being the energy tensor of the gravitational field itself. Forming the divergence of (8) and requiring that the terms of the third differential order cancel among themselves — since $\mathfrak{t}$ contains no such terms — one obtains a single condition, derived in Appendix C:

$$a_5 = 0. \tag{14}$$

Of the 16 candidates surviving Sec. IV, this eliminates 12, marked II in Table II.

VI. THE SURVIVING CANDIDATES

Four candidates remain: $K_{18}, K_{21}, K_{28}, K_{31}$, distinguished only by the value of a_2 (and hence of $a_6 = -a_2$). Table I compares them.

The integration is elementary in each case and is not reproduced. Three of the four fail at once. There remains K_{28},

$$E_{\mu\nu} = S^{(1)}_{\mu\nu} + \tfrac{1}{2}S^{(2)}_{\mu\nu} - \tfrac{1}{2}S^{(4)}_{\mu\nu} - \tfrac{1}{2}S^{(6)}_{\mu\nu}. \tag{15}$$

Upon reduction, (15) is found to be identical with the left-hand member of the field equations obtained in

Ref. [5] by a wholly different route, namely from the requirement that the force density exerted by the field on matter be expressible as a divergence. That two arguments of so different a character — an exhaustive classification on the one hand, the law of conservation on the other — should converge upon the same expression is, in our judgement, the strongest evidence yet available for the correctness of those equations.

VII. THE IMPOSSIBILITY OF GENERAL COVARIANCE

It remains to justify the restriction of Sec. II to linear substitutions.

Under an arbitrary substitution $x'^\mu = f^\mu(x)$ each of the six structures $S^{(n)}$ acquires a term containing the second derivatives $\partial^2 x^\alpha/\partial x'^\mu \partial x'^\nu$. For $E_{\mu\nu}$ to transform as a tensor these terms must cancel among themselves. Collecting them, the condition of cancellation is the system

$$a_1+a_3+2a_4=0, \quad a_2+a_5=0, \quad a_3=0, \quad a_2=0, \tag{16}$$

in which the first is satisfied identically by (9). The last two, taken with (14), would give $a_2=a_3=a_5=0$, that is, the candidate

$$E_{\mu\nu} = S^{(1)}_{\mu\nu} - \tfrac{1}{2} S^{(4)}_{\mu\nu}, \tag{17}$$

which is K_{23} of Table II; and K_{23} has been eliminated in Sec. IV, since with $a_2=0$ the field of a mass point does not reduce to the Newtonian potential at large distance but contains an additional logarithmic term.

Thus no expression satisfying the conditions of Sec. II is generally covariant. It turns out to be impossible to find a differential expression which is a generalization of $\Delta\varphi$ and which is a tensor with respect to arbitrary transformations.

We are conscious of how unwelcome this conclusion is. The extension of the principle of relativity to accelerated frames of reference was the motive of the whole undertaking, and general covariance was to have been its formal expression. The classification shows that the two cannot be had together. Since the requirement of Sec. II C rests upon the equality of inertial and gravitational mass, which is an experimental result of the highest accuracy, and the requirement of general covariance rests upon an aesthetic preference, it is the latter which must yield.

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Appendix A: The weight relations

Under $x'^\mu = \Lambda^\mu{}_\nu x^\nu$ with $\det\Lambda = \lambda$, the determinant g acquires the factor λ^{-2} and $\ln\sqrt{-g}$ the additive constant $-\ln\lambda$, which is annihilated by the derivatives in $S^{(2)}$ and $S^{(5)}$ but not by the contraction in $S^{(4)}$ and $S^{(6)}$. Requiring that the sum (8) carry weight zero gives (9).

Appendix B: The identities used in the reduction

The 28 equivalences referred to in Sec. III follow from

$$S^{(3)}_{\mu\nu} \equiv 2S^{(2)}_{\mu\nu} + g_{\mu\nu}S^{(5)} - S^{(6)}_{\mu\nu} + \partial_\mu\partial_\nu\big(g^{\alpha\beta}g_{\alpha\beta}\big), \tag{B1}$$

$$S^{(5)}_{\mu\nu} \equiv g_{\mu\nu}\, g^{\alpha\beta} S^{(2)}_{\alpha\beta} - \tfrac{1}{2} g_{\mu\nu} g^{\alpha\beta} S^{(6)}_{\alpha\beta}, \tag{B2}$$

which hold for any $g_{\mu\nu}$ of vanishing determinant derivative. Expressions differing only by multiples of these are counted once.

Appendix C: The condition (14)

Forming $\partial^\nu E_{\mu\nu}$ and retaining the terms of the third differential order,

$$\partial^\nu E_{\mu\nu}\big|_3 = a_5\left(\partial_\mu g^{\alpha\beta}\partial_\alpha\partial_\beta - \partial_\mu\Box\right)\ln\sqrt{-g}, \tag{C1}$$

the contributions of $S^{(1)}, S^{(2)}, S^{(4)}, S^{(6)}$ cancelling in pairs by (9). Since $\mathfrak{t}_\mu^\nu$ is quadratic in the first derivatives and contains no third derivatives, (13) requires the above to vanish identically, whence (14).

Appendix D: The complete list of admissible expressions

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TABLE II. The 47 distinct candidates. Coefficients refer to (8) with $a_1 = 1$; a_4 and a_6 are fixed by (9). Column "El." gives the round of elimination: I, Newtonian limit (Sec. IV); II, conservation (Sec. V); a dash marks the survivors.

	a_2	a_3	a_4	a_5	a_6	El.
K1	-1	-1	0	$-\frac{1}{2}$	1	I
K2	-1	-1	0	0	1	I
K3	-1	-1	0	$\frac{1}{2}$	1	I
K4	$-\frac{1}{2}$	-1	0	-1	$\frac{1}{2}$	I
K5	$-\frac{1}{2}$	-1	0	0	$\frac{1}{2}$	I
K6	$-\frac{1}{2}$	-1	0	1	$\frac{1}{2}$	I
K7	0	-1	0	-1	0	I
K8	0	-1	0	$-\frac{1}{2}$	0	I
K9	0	-1	0	$\frac{1}{2}$	0	I
K10	0	-1	0	1	0	I
K11	$\frac{1}{2}$	-1	0	-1	$-\frac{1}{2}$	I
K12	$\frac{1}{2}$	-1	0	0	$-\frac{1}{2}$	I
K13	$\frac{1}{2}$	-1	0	1	$-\frac{1}{2}$	I
K14	1	-1	0	$-\frac{1}{2}$	-1	I
K15	1	-1	0	0	-1	I
K16	1	-1	0	$\frac{1}{2}$	-1	I
K17	-1	0	$-\frac{1}{2}$	$-\frac{1}{2}$	1	II
K18	-1	0	$-\frac{1}{2}$	0	1	–
K19	-1	0	$-\frac{1}{2}$	$\frac{1}{2}$	1	II
K20	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$\frac{1}{2}$	II
K21	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	–
K22	$-\frac{1}{2}$	0	$-\frac{1}{2}$	1	$\frac{1}{2}$	II
K23	0	0	$-\frac{1}{2}$	-1	0	II
K24	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	0	II
K25	0	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	II
K26	0	0	$-\frac{1}{2}$	1	0	II
K27	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	II
K28	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	$-\frac{1}{2}$	–
K29	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	$-\frac{1}{2}$	II
K30	1	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	II
K31	1	0	$-\frac{1}{2}$	0	-1	–
K32	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	-1	II
K33	-1	1	-1	$-\frac{1}{2}$	1	I
K34	-1	1	-1	0	1	I
K35	-1	1	-1	$\frac{1}{2}$	1	I
K36	$-\frac{1}{2}$	1	-1	-1	$\frac{1}{2}$	I
K37	$-\frac{1}{2}$	1	-1	0	$\frac{1}{2}$	I
K38	$-\frac{1}{2}$	1	-1	1	$\frac{1}{2}$	I
K39	0	1	-1	-1	0	I
K40	0	1	-1	$-\frac{1}{2}$	0	I
K41	0	1	-1	$\frac{1}{2}$	0	I
K42	0	1	-1	1	0	I
K43	$\frac{1}{2}$	1	-1	-1	$-\frac{1}{2}$	I
K44	$\frac{1}{2}$	1	-1	0	$-\frac{1}{2}$	I
K45	$\frac{1}{2}$	1	-1	1	$-\frac{1}{2}$	I
K46	1	1	-1	$-\frac{1}{2}$	-1	I
K47	1	1	-1	0	-1	I